

Hub Motor

Technical Data Sheet



Overview

The Mecwin 12-Inch Hub Motor is a robust and efficient BLDC motor designed for modern electric mobility solutions. With its integrated wheel design, it provides smooth performance, high torque, and low maintenance, making it perfect for electric scooters, e-bikes, and lightweight EVs. Built using advanced technology and durable materials, it ensures reliable operation, energy efficiency, and long service life—demonstrating Mecwin's commitment to innovation and sustainable mobility.



Motor Specification :

Motor Specification			
Parameters	Specifications	Parameters	Specifications
Operating Voltage	72 V	Rated Torque	23 Nm
Rated Power	2.0 kW	Peak Torque	132 Nm
Peak Power	2.7 kW	Max Speed	884 Rpm
Max. DC Current	50 A		

Motor Specification			
Parameters	Specifications	Parameters	Specifications
Outer Diameter	332 mm	Break Type	130 mm Drum
Shaft Length	275 mm	Swing Arm Distance	192.5 mm
IP Level	IP67	Application	2 wheeler

Performance Details :

Parameter	DC Voltage	DC Current	RMS Line to Line Voltage	RMS Line Current	Input Power	Torque	Speed	Output power	System Efficiency
	V	A	W	V	A	N-m	RPM	W	%
No Load	72	6.92	55.9	7.4	498.37	0	884	0.35	0.07
Max Torque	72	47.19	12.63	119.96	3397.47	132	94.08	1307.81	38.49
Max Input Power	72	48.63	14.58	111.67	3501.5	123	136.08	1752.69	50.06
Max Output Power	72	49.37	47.9	48.44	3055.03	40	619.45	2700	88.379
Max Efficiency	72	49.37	47.9	48.44	3055.03	40	619.45	2700	88.379

For More Information, Contact Mecwin :

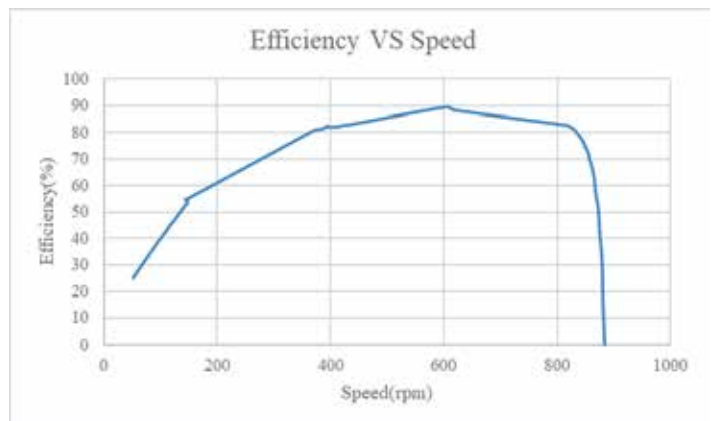
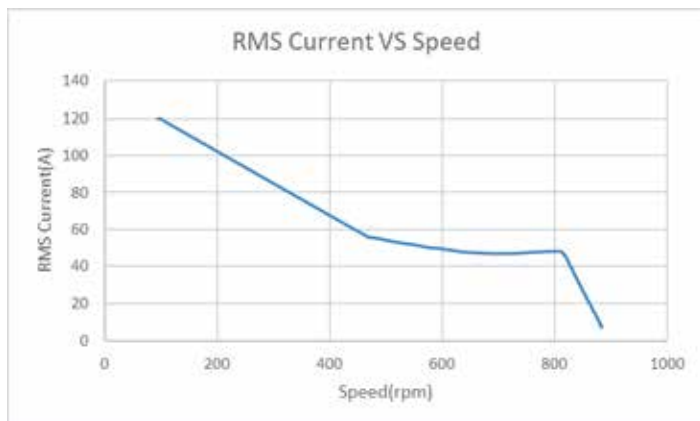
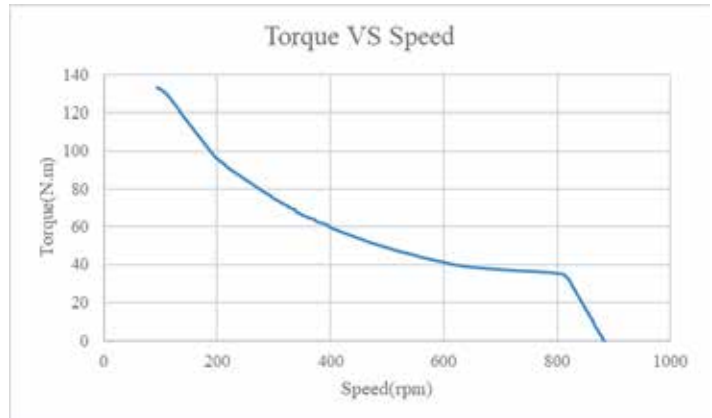
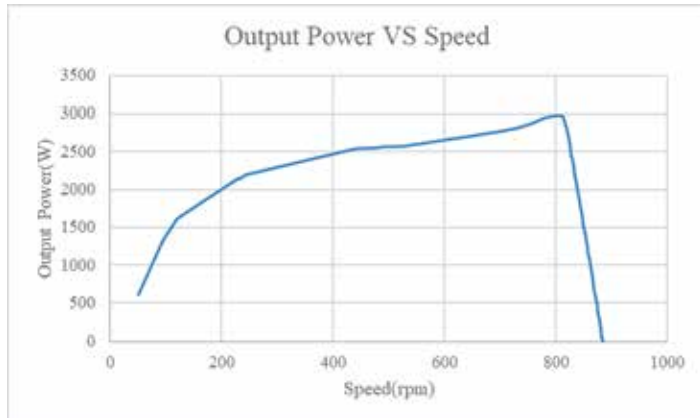
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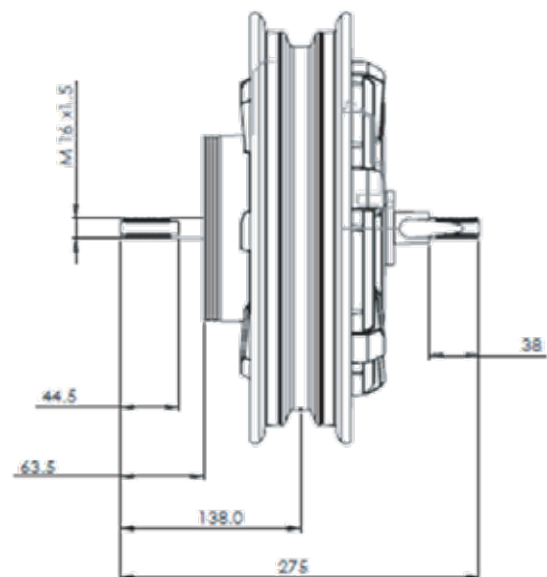
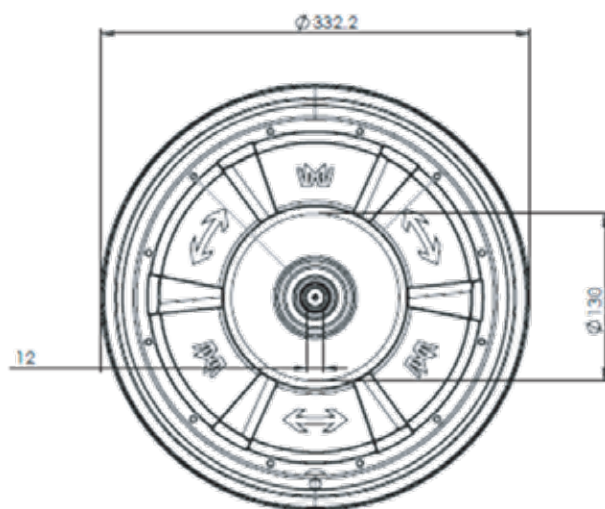
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Performance Curves :



Mechanical 2D Diagram :



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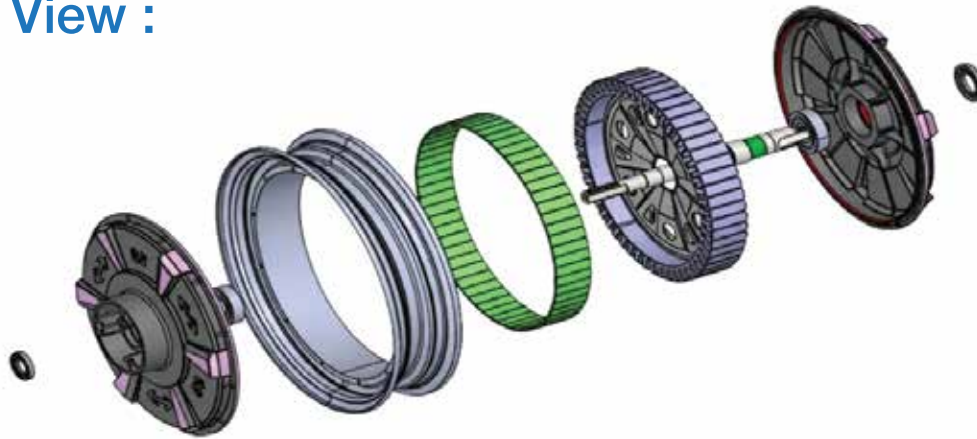
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Exploded View :



Technical Data :

DC Voltage	DC Current	RMS Line to Line Voltage	RMS Line Current	Input Power	Torque	Speed	Output power	System Efficiency
V	A	W	V	A	N-m	RPM	W	%
72	6.92	55.9	7.4	498.37	0	884	0.35	0.07
72	8.13	55.94	8.43	585.32	1	881	92.42	15.79
72	9.4	55.96	9.54	677.03	2	880	184.26	27.22
72	10.71	56.02	10.73	770.88	3	878	276.08	35.81
72	11.93	56.05	11.86	859.15	4	875	366.42	42.65
72	13.22	56.09	13.09	952.03	5	874	457.69	48.07
72	14.49	56.14	14.29	1043.47	6	872	548.02	52.52
72	15.72	56.18	15.39	1131.85	7	869	636.96	56.28
72	16.99	56.22	16.54	1222.96	8	867	726.33	59.39
72	18.21	56.28	17.65	1311.33	9	866	816.17	62.24
72	19.44	56.31	18.78	1399.59	10	864	904.69	64.64
72	20.7	56.36	19.95	1490.24	11	862	992.61	66.61
72	21.95	56.42	21.12	1580.58	12	860	1080.93	68.39
72	23.15	56.44	22.23	1666.78	13	857	1167.31	70.03
72	24.38	56.48	23.4	1755.5	14	856	1254.15	71.44
72	25.59	56.53	24.53	1842.76	15	854	1341.39	72.79
72	26.82	56.58	25.71	1930.7	16	852	1426.64	73.89
72	28.07	56.64	26.94	2020.93	17	849	1511.55	74.79
72	29.28	56.67	28.07	2108.45	18	848	1597.86	75.78
72	30.54	56.73	29.31	2198.86	19	846	1682.86	76.53
72	31.76	56.75	30.44	2286.96	20	844	1766.77	77.25
72	32.97	56.8	31.57	2373.55	21	842	1851.29	78
72	34.19	56.87	32.77	2461.31	22	840	1934.58	78.6
72	35.45	56.91	33.97	2552.31	23	838	2017.37	79.04
72	36.62	56.95	35.12	2636.92	24	836	2100.65	79.66
72	37.84	57	36.28	2724.38	25	834	2183.7	80.15

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72	39.03	57.04	37.45	2809.93	26	832	2266.12	80.65
72	40.24	57.08	38.61	2897.14	27	831	2348.33	81.06
72	41.45	57.12	39.82	2984.48	28	828	2428.78	81.38
72	42.67	57.18	41	3071.75	29	826	2508.25	81.66
72	43.89	57.2	42.16	3160.47	30	825	2590.57	81.97
72	45.11	57.25	43.3	3248.02	31	823	2670.12	82.21
72	46.34	57.31	44.49	3337.1	32	821	2750.69	82.43
72	47.49	57.32	45.69	3418.27	33	818	2828.05	82.73
72	48.59	57.35	46.84	3497.55	34	815	2901.74	82.96
72	49.57	57.35	47.9	3568.57	35	810	2970.38	83.24
72	49.69	57.04	47.85	3538.85	36	782	2949.99	83.36
72	49.65	55.27	46.83	3354.02	37	723	2800.28	83.49
72	49.48	52.67	46.95	3174.86	38	677	2695.62	84.91
72	49.46	49.85	47.55	3070.46	39	641	2619.52	85.31
72	49.37	47.9	48.44	3055.03	40	619	2700	88.38
72	49.36	46.53	49.13	3048.13	41	604	2730	89.56
72	49.39	45.19	49.75	3043.78	42	589	2675	87.88
72	49.39	43.77	50.1	3040.44	43	574	2582.6	84.94
72	49.39	42.58	50.98	3039.26	44	560	2580.64	84.91
72	49.4	41.58	51.61	3062.44	45	549	2585.33	84.42
72	49.38	40.55	52.18	3045.88	46	536	2582.96	84.8
72	49.38	39.37	52.73	3044.1	47	522	2569.88	84.42
72	49.38	38.41	53.36	3045.49	48	511	2566.8	84.28
72	49.37	37.51	54.09	3044.8	49	500	2565.65	84.26
72	49.41	36.64	54.79	3049.18	50	489	2562.54	84.04
72	49.41	35.77	55.5	3050.49	51	478	2552.05	83.66
72	49.37	34.92	55.99	3041.65	52	467	2544.09	83.64
72	49.37	34.27	56.96	3042.34	53	459	2547.62	83.74
72	49.38	33.56	57.7	3043.82	54	449	2540.19	83.45
72	49.37	32.89	58.74	3051.5	55	440	2536.92	83.14
72	49.37	32.21	59.36	3051.07	56	431	2529.17	82.89
72	49.36	31.53	59.88	3046.25	57	422	2521.69	82.78
72	49.36	30.84	60.59	3049.5	58	413	2508.65	82.26
72	49.35	30.23	61.26	3047.89	59	405	2501.52	82.07
72	49.33	29.7	62.14	3042.81	60	398	2498.13	82.1
72	49.35	29.5	63.19	3055.21	61	395	2520.44	82.5
72	49.37	28.85	63.44	3065	62	384	2493.72	81.36
72	49.34	28.17	64.31	3053.37	63	375	2476.49	81.11
72	49.36	27.82	65.01	3073.83	64	371	2486.75	80.9
72	49.33	27.17	65.48	3063.22	65	360	2453.05	80.08
72	49.32	26.53	66.42	3055.12	66	352	2433.18	79.64
72	49.34	26.11	67.05	3054.01	67	346	2429.07	79.54
72	49.32	25.61	67.78	3048.2	68	339	2417.46	79.31
72	49.31	25.49	68.78	3050.66	69	338	2438.87	79.95
72	49.32	25.06	69.54	3073.45	70	329	2415.07	78.58

72	49.36	24.79	70.28	3055.77	71	325	2414.13	79
72	49.38	24.28	71.15	3058.01	72	318	2395.02	78.32
72	49.37	23.93	71.67	3065.72	73	312	2381.89	77.69
72	49.32	23.56	72.61	3071.77	74	306	2369.56	77.14
72	49.28	23.09	73.23	3060.74	75	299	2350.21	76.79
72	49.28	22.88	74.08	3079.47	76	296	2352.62	76.4
72	49.25	22.54	74.57	3085	77	291	2343.61	75.97
72	49.26	22.19	75.45	3043.01	78	285	2324.84	76.4
72	49.28	21.69	76.39	3042.13	79	277	2287.81	75.2
72	49.31	21.41	77.04	3068.57	80	273	2282.89	74.4
72	49.33	20.74	77.71	3077.37	81	267	2264.48	73.58
72	49.3	20.43	78.27	3054.28	82	264	2263.32	74.1
72	49.25	20.25	78.92	3057.25	83	260	2255.84	73.79
72	49.2	20.36	80.04	3055.37	84	260	2286.56	74.84
72	49.27	19.73	80.29	3056.95	85	250	2222.67	72.71
72	49.26	19.29	81.07	3058.04	86	243	2192.02	71.68
72	49.28	18.96	81.84	3087.4	87	238	2166.07	70.16
72	49.28	18.7	83.51	3056.39	88	232	2138.18	69.96
72	49.27	18.55	83.59	3056.86	89	230	2144.47	70.15
72	49.36	18.21	83.79	3070.2	90	225	2116.27	68.93
72	49.3	17.98	84.57	3051.19	91	220	2094.77	68.65
72	49.25	17.81	85.63	3047.19	92	216	2082.16	68.33
72	49.22	17.57	85.95	3046.1	93	213	2072.96	68.05
72	49.24	17.41	86.94	3074.09	94	209	2059.6	67
72	49.22	17.02	87.34	3072.31	95	204	2026.89	65.97
72	49.16	16.77	87.44	3063.44	96	201	2019.48	65.92
72	49.17	16.55	88.1	3074.03	97	196	1991.52	64.79
72	49.25	16.46	89.54	3062.06	98	193	1982.75	64.75
72	49.23	16.37	90.96	3078.84	99	191	1977.04	64.21
72	49.19	15.91	91.62	3086.46	100	183	1918.06	62.14
72	49.19	15.85	93.13	3045.43	101	182	1921.58	63.1
72	49.23	15.58	94.06	3045.26	102	176	1878.15	61.67
72	49.22	15.33	94.92	3035.68	103	171	1843.43	60.73
72	49.22	15.15	95.63	3035.73	104	167	1823.14	60.06
72	49.22	14.99	96.41	3035.69	105	164	1808.09	59.56
72	49.25	14.84	96.99	3037.16	106	161	1785.33	58.78
72	49.26	14.65	97.87	3038.99	107	158	1764.82	58.07
72	49.23	14.49	98.72	3042.78	108	154	1740.63	57.21
72	49.25	14.41	99.54	3099.35	109	152	1732.46	55.9
72	49.23	14.15	100.5	3081.27	110	147	1692.84	54.94

72	49.22	13.97	101.14	3049.82	111	144	1668.65	54.71
72	49.18	13.96	102.03	3056	112	143	1677.56	54.89
72	49.16	14.1	103.07	3104.61	113	144	1699.65	54.75
72	49.12	14.22	104.05	3164.07	114	145	1731.51	54.72
72	49.1	14.37	105.17	3213.52	115	146	1755.49	54.63
72	49.05	14.47	105.92	3268.43	116	146	1777.47	54.38
72	49.05	14.61	106.97	3329.33	117	147	1805.91	54.24
72	49.05	14.76	107.82	3388.39	118	148	1824.96	53.86
72	49.01	15.01	108.83	3445.98	119	148	1843.08	53.48
72	48.97	15.12	109.47	3490.28	120	148	1854.21	53.13
72	48.81	14.98	110.43	3507.48	121	144	1829.84	52.17
72	48.75	14.76	111.27	3509.7	122	140	1792.53	51.07
72	48.63	14.58	111.67	3501.5	123	136	1752.69	50.06
72	48.54	14.37	112.15	3494.81	124	132	1718.32	49.17
72	48.47	14.21	113.01	3490.07	125	128	1681.16	48.17
72	48.44	14.01	113.88	3487.66	126	125	1644.69	47.16
72	48.4	13.86	114.47	3484.77	127	121	1614.35	46.33
72	48.37	13.66	115.49	3482.33	128	117	1571.28	45.12
72	48.36	13.53	116.8	3481.44	129	113	1531.99	44
72	48.32	13.35	117.76	3478.86	130	110	1490.95	42.86
72	48.22	13.16	118.76	3472.17	131	105	1443.33	41.57
72	47.9	12.95	119.6	3448.95	132	101	1391.29	40.34
72	47.19	12.63	119.96	3397.47	133	94	1307.81	38.49
72	34.27	9.32	102.22	2412.57	114	51	605.21	25.09